

MONOCLOX 250MG, 500MG POWDER FOR SOLUTION FOR INJECTION OR INFUSION
SUMMARY OF PRODUCT CHARACTERISTICS

1. NAME OF MEDICINAL PRODUCT

MONOCLOX 250mg powder for solution for injection or infusion

MONOCLOX 500mg powder for solution for injection or infusion

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

Each vial contains dry powder cloxacillin sodium for solution for injection or infusion equivalent to 250mg and 500mg cloxacillin, respectively.

For full list of excipients, see section 6.1

3. PHARMACEUTICAL FORM

Powder for solution for injection or infusion

A white or almost white crystalline powder

4. CLINICAL PARTICULARS

4.1 Therapeutic indications

- Cloxacillin is indicated for the treatment of sensitive staphylococcal infections: respiratory infections, otorhinolaryngologic infections, renal infections, urogenital infections, neuro-meningeal infections, bone and joint infections, endocarditis and treatment of skin infections caused by sensitive staphylococci and/or streptococci

4.2 Posology and method of administration

Monoclox injection is for intramuscular or intravenous administration.

Preparation of the solution for i.m administration:

The content of each vial of Monoclox 250mg should be dissolved in 1.5ml of water for injections and shake well until complete dissolution. The solution should not be mixed with another product. Any solution left pending must be discarded.

The content of each vial of Monoclox 500mg should be dissolved in 2.5ml of water for injections and shake well until complete dissolution. The solution should not be mixed with another product. Any solution left pending must be discarded.

DO NOT USE INTRAMUSCULAR SOLVENT BY INTRAVENOUS ROUTE

Preparation for i.v administration: For intravenous injection each vial of Monoclox 250mg and 500mg respectively should be dissolved in 10-20ml water for injection and shake well.

For slow intravenous infusion Monoclox can be further diluted in 100ml of an infusion fluid. Infuse the solution slowly over a period of 60 minutes.

Dosage (in patients with normal renal and hepatic function):

- Treatment of infections:

Adult:

- 1-2g every 6 hours
- For more severe infections (e.g. meningitis, endocarditis) 2g every 4 hours

Child:

- neonates 7 days of age and younger, weighing less than 2kg: 25mg/kg intravenously every 12 hours
- neonates 7 days to 28 days of age, weighing less than 2kg, or neonates 7 days of age or younger weighing 2kg or more: 25mg/kg intravenously every 8 hours
- neonates 7 days to 28 days of age weighing 2kg or more: 25mg/kg intravenously every 6 hours. For treatment of meningitis, the above doses should be doubled and given intravenously
- children from 1 month of age, weighing less than 20kg: 50 to 100mg/kg daily (maximum 4g) divided

every 6 hours; intravenously or intramuscularly. For severe infection, up to 200mg/kg daily (maximum 12g) in divided dose may be given intravenously.

- children from 1 month of age, weighing 20kg or more: as for adults.

4.3 Contraindications

Monoclox is contraindicated to patients who are hypersensitive to cloxacillin or other penicillins, cephalosporins or penicillamine.

Monoclox should not be administered by sub-conjunctival injection.

The administration of cloxacillin in newborns is contraindicated.

4.4 Special warnings and precautions for use

Cloxacillin should be given with caution to patients who have had previous hypersensitivity reactions (anaphylactic reactions) to the cephalosporins, penicillins or other drugs. If serious acute hypersensitivity or allergic reactions take place, patients should be treated with pressor amines, antihistamines, corticosteroids and/or other emergency measures. Hypersensitivity reactions can sometimes be quite severe or even fatal. Even though anaphylaxis is more likely to happen after parenteral administration they have also been induced by oral administration as well.

Extensive use of antibiotics may encourage the overgrowth of non-susceptible organisms, including fungi. In such a case, appropriate measures should be taken. It is also advised to evaluate, periodically, the renal, hepatic and hematopoietic systems.

In streptococcal infections, treatment must be adequate to eliminate the organism (10 days minimum); otherwise the sequelae of streptococcal disease may occur. The cultures should be taken when the treatment has finished so to determine whether streptococci have been eradicated.

4.5 Interactions with other medicinal products and other forms of interaction

Combinations not recommended

Methotrexate

Increased effects and haematological toxicity of methotrexate: inhibition of renal tubular secretion of methotrexate by penicillins.

Specific problems of the imbalance of the INR

Many cases of increased activity of oral anticoagulants have been reported in patients receiving antibiotics. The marked infectious or inflammatory context, the age and the general state of the patient appear as risk factors. In these circumstances, it appears difficult to distinguish between the infectious pathology and its treatment in the occurrence of the imbalance of the INR. However, some classes of antibiotics are more involved, including fluoroquinolones, macrolides, cyclins, cotrimoxazole and some cephalosporins.

4.6 Pregnancy and lactation

Pregnancy category: B

In clinical practice, analysis of a large number of exposed pregnancies did not appear to reveal any malformative or foetotoxic effect of penicillins. Therefore, cloxacillin may be prescribed during pregnancy if necessary.

Lactation

Because cloxacillin passes into breast milk, possible suspension of breastfeeding should be envisaged.

4.7 Effects on the ability to drive and use machinery

Cloxacillin does not influence the patient's ability to drive or operate machinery.

4.8 Undesirable effects

Immune system disorders

Urticaria, Quincke's edema, exceptionally anaphylactic shock, eosinophilia (see section 4.4).

Skin and subcutaneous tissue disorders

Maculopapular rash of allergic origin or not. DRESS syndrome (drug hypersensitivity syndrome with eosinophilia and systemic symptoms).

Isolated cases of erythroderma and severe bullous toxin (erythema multiforme, Stevens Johnson syndrome, Lyell syndrome).

Gastrointestinal disorders

Nausea, vomiting, diarrhea.

Rare cases of pseudomembranous colitis have been reported (see section 4.4).

Hepatobiliary disorders

Rare and moderate increase in transaminases (ASAT and ALAT), exceptionally cholestatic hepatitis.

Nervous system disorders

Betalactamines exhibit risk for encephalopathy (confusion, disturbances of consciousness, epilepsy or abnormal movements) and, especially, in case of overdose or renal failure.

Renal and urinary disorders

Acute immunoallergic interstitial nephritis.

Blood and lymphatic system disorders

Reversible haematological disorders: anemia, thrombocytopenia, leukopenia, neutropenia and agranulocytosis.

General disorders and administration site conditions

Fever.

Reporting of suspected adverse reactions

Reporting suspected adverse reactions after authorisation of the medicinal product is important. It allows continued monitoring of the benefit/risk balance of the medicinal product. Healthcare professionals are asked to report any suspected adverse reactions via the national reporting system.

4.9 Overdosage

Betalactamines exhibit risk for encephalopathy (confusion, disturbances of consciousness, epilepsy or abnormal movements) and, especially, in case of overdose or renal failure.

Manifestations of overdose, neuropsychic, renal and digestive, have been reported with penicillins M.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

Pharmacotherapeutic group: Beta-lactamase resistant penicillin's, ATC code: J01CF02.

Cloxacillin is an antibiotic of the beta-lactam family of the penicillins M group.

Spectrum of antibacterial activity

Breakpoints

The critical minimum inhibitory concentrations (MICs) established by the European Committee on Antimicrobial Susceptibility Testing (EUCAST) are presented below.

<i>Critical concentrations established by EUCAST for cloxacillin (2010-04-27, v.1.1)</i>		
<i>Organism</i>	<i>Susceptible (S) (mg/l)</i>	<i>Resistant (R) (mg/l)</i>
<i>Staphylococcus aureus</i>	≤ 2	>2
<i>Staphylococcus lugdunensis</i>	≤ 2	>2
<i>Staphylococci coagulase negative</i>	≤ 0.25	>0.25

The prevalence of acquired resistance may vary with geography and time for some species. It is therefore

useful to have information on the prevalence of local resistance, especially for the treatment of severe infections. If necessary, it is desirable to obtain a specialized opinion mainly when the interest of the drug in certain infections can be questioned because of the level of prevalence of the local resistance.

<i>Category</i>
<i>Commonly susceptible species</i> <u>Gram-positive aerobes</u> <i>Streptococcus pyogenes</i> <u>Anaerobes</u> <i>Clostridium perfringes</i>
<i>Inconstantly sensitive species (Acquired resistance $\geq 10\%$)</i> <u>Gram-positive aerobes</u> <i>Staphylococcus aureus</i> (1) <i>Staphylococci</i> coagulase negative (+)
(+) The prevalence of bacterial resistance is $\geq 50\%$ in France. (1) The frequency of resistance to methicillin is approximately 20 to 30% in <i>Staphylococcus aureus</i> and occurs mainly in hospital.

5.2 Pharmacokinetic properties

Absorption

Cloxacillin is stable in gastric environment. It is well absorbed by the digestive mucous (70%).

Distribution

- After oral administration, maximum blood concentrations are obtained after 1 hour and are proportional to the administered dose. They are in order of 9mg/l for one dose of 500mg
- After intramuscular injection, maximum blood concentrations are obtained after about 45 minutes. In adults, they are 26mg/l for one dose of 1g, 15mg/l for one dose of 500mg and 8mg/l for one dose of 250mg.
- After intravenous injection of 2g perfused for 20 minutes, the value of serum peak obtained at the end of the perfusion is 280mg/l.
- Half-life is in the order of 45 minutes in patients with normal renal functions.
- Plasma protein bounding is in the order of 90%.
- The products diffuses the foetal blood, synovial liquid and bone tissue into the amniotic fluid.

Biotransformation

Cloxacillin is metabolised to a limited extent.

Excretion

After oral administration, the non-absorbed fraction is eliminated by the intestinal route in an inactive form. The absorbed fraction is eliminated in an inactive form essentially through urine and for 10% through the bile duct.

After administration by injection, the elimination is:

- Urinary, in an active form, in 6 hours, approximately 70 to 80% of the injected dose
- Biliary, in an active form, 20 to 30% of the injected dose

5.3 Preclinical safety data

Animal studies showed no teratogenesis, carcinogenesis or mutagenesis. There have been no relevant studies with humans.

6. PHARMACEUTICAL DATA

6.1 List of excipients: None

6.2 Incompatibilities

Cloxacillin is incompatible with solutions of amino acids, lipid emulsions and blood for transfusion

6.3 Shelf life

36 months (3 years)

Reconstituted solution: the product should be used immediately.

6.4 Special precautions for storage

Store below 25°C

6.5 Nature and contents of container

Clear type III glass vials, nominal capacity 8ml sealed with 20mm rubber stopper, aluminium caps with plastic flip-off seal, labelled and packed in a card carton

Box of 100 vials is available.

6.6 Special precautions for disposal: None

7. MANUFACTURER

MEDOCHÉMIE LTD. (Fast East) Ltd., (Aseptic Penicillin facility), No. 10, 12 and 16 Street 27, Vietnam
Singapore Industrial Park II A, Vinh Tan Commune, Tan Uyen, Vietnam

8. PRODUCT REGISTRATION HOLDER

KOMEDIC SDN BHD, 4, Jalan PJS 11/14, Bandar Sunway, 46150 Petaling Jaya Selangor

9. DATE OF REVISION

03/2025